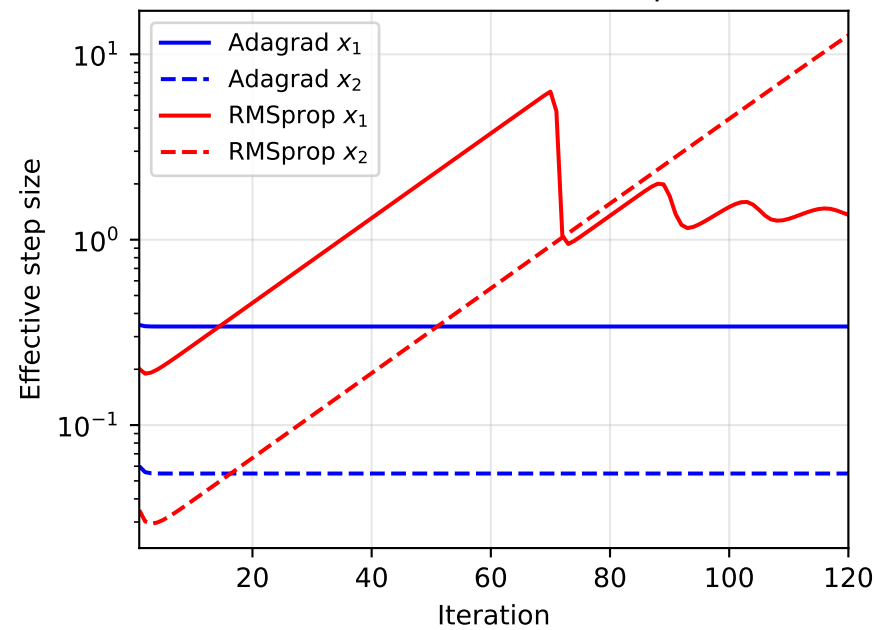
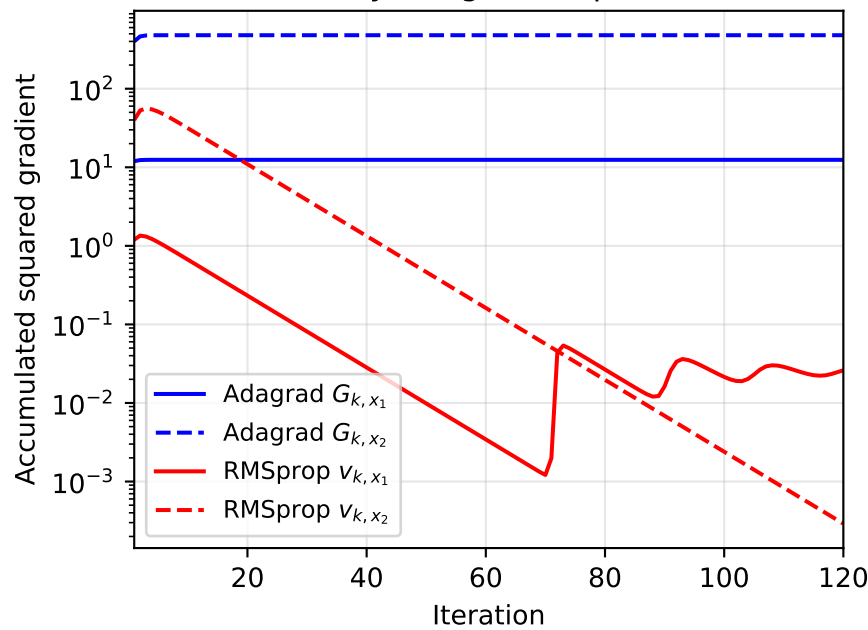


Adagrad vs RMSprop: Anisotropic Problem (Benchmark B, $\kappa \approx 5$)

Per-coordinate step sizes
(Benchmark B, anisotropic)



G_k (Adagrad) vs v_k (RMSprop)
shows why Adagrad step vanishes



Convergence on Benchmark B
(Adagrad eventually stalls)

